

Notes: Capatina & Keane (JPE, 2026)
Health Shocks, Health Insurance, Human Capital, and the Dynamics of
Earnings and Health

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1 Big picture

- **Question.** How do health shocks affect health, labor supply, human capital, earnings, earnings inequality, and the value of health insurance over the life cycle?
- **Core mechanism.** Health shocks affect earnings through four connected channels: immediate lost work time, lower health productivity, reduced future human capital accumulation, and behavioral adjustments to health risk and the social insurance environment.
- **Central modeling idea.** The paper embeds health shocks and health insurance into a dynamic life-cycle model with employment offers, tied wage-hours-insurance packages, medical treatment choices, out-of-pocket costs, free care, bill default, assets, work experience, and endogenous public transfers.
- **Why insurance is multidimensional.** Insurance does not only smooth consumption by reducing OOP medical costs. It also grants access to treatment, changes the likelihood of treating shocks, improves future health, and alters labor-supply incentives.
- **Main quantitative result on health shocks.** A severe health shock at working ages causes large and persistent earnings losses, especially for lower-educated men. The losses are amplified by reductions in human capital accumulation and slow re-entry into full-time employment.
- **Main inequality result.** Health shocks explain roughly 12–13% of lifetime earnings inequality for White men and more within low-education groups, because low-skill workers are both more exposed to health-related labor-market exits and more affected by incentives to rely on transfers.
- **Main policy result.** Providing public insurance to men without employer-sponsored health insurance greatly increases treatment access and medical spending, but it also reduces unpaid bills, Medicaid/free-care spending, social-insurance reliance, and strengthens labor-supply incentives.
- **Key policy interpretation.** Universal insurance for the uninsured can be more labor-supply-friendly than means-tested Medicaid because it does not require workers to remain low income to preserve health coverage.
- **Economic takeaway.** Health is both a consumption-risk object and a human-capital/labor-supply state variable. Policies that improve access to treatment can raise lifetime earnings by preventing health shocks from turning into persistent labor-market detachment.

2 Incremental contribution

- **Integrates two health literatures.** The paper combines the Grossman-style view of health spending as voluntary investment with the De Nardi–French–Jones style view of health shocks as costly medical events. Treatment is a choice, not an automatic expenditure.
- **Adds endogenous treatment to a life-cycle earnings model.** Prior life-cycle models often treat medical spending or health transitions as exogenous. Here, agents decide whether to treat shocks, whether to pay OOP costs, whether to default, and how treatment affects future health.
- **Connects health shocks to human capital.** The paper makes work experience and labor-market attachment part of the health-shock transmission mechanism. A health shock can reduce current work and thereby reduce future wage offers through lower human capital.
- **Models tied job offers.** Wage offers, hours, and employer-sponsored insurance arrive as bundled employment opportunities. This lets health shocks affect employment not only through preferences and productivity but also through future offer probabilities.
- **Separates cost access and treatment access.** Insurance changes OOP costs, but it also changes whether a worker can obtain care. The paper shows that access to treatment can be more important than the OOP cost channel.

- **Explains observed health gradients without direct education effects in health production.** The health transition equation does not require education to enter directly. Observed education-health correlations arise because education, latent health type, treatment choices, income, and insurance are jointly selected.
- **Quantifies inequality channels.** The paper decomposes the effect of health shocks on lifetime earnings into labor supply, human capital, health productivity, and behavioral effects, showing that human capital and behavior are central for low-skill workers.
- **Evaluates public insurance with budget feedbacks.** The counterfactual does not only count medical costs. It also includes tax revenues, social-insurance spending, Medicaid/free-care savings, unpaid bills, Medicare/Social Security effects, and welfare under balanced-budget financing.
- **Extends the analysis to racial minorities.** The model is recalibrated for Black and Hispanic men with high school or less education to assess whether the health-shock mechanisms are similar across groups.

3 Data and measurement

3.1 Data sources

- **MEPS.** The Medical Expenditure Panel Survey provides information on health status, health shocks, insurance coverage, medical spending, and treatment behavior.
- **CPS.** The Current Population Survey is used to discipline employment, wages, insurance, and earnings patterns across ages and education groups.
- **HRS.** The Health and Retirement Study helps discipline later-life assets, health, mortality, and health-related labor-market transitions.
- **PSID.** The Panel Study of Income Dynamics contributes life-cycle labor-market and asset dynamics used in calibration.

Interpretation: The empirical strategy is structural calibration, not a single reduced-form research design. Each data set anchors a different part of the life-cycle model: health shocks and spending, job offers and earnings, assets, mortality, and later-life transitions.

3.2 Health states, health shocks, and risk factors

- Functional health is summarized by a discrete state H_t , typically interpreted as good, fair, or poor.
- Risk factors R_t , such as predictable health risks, affect the likelihood of future shocks.
- The model distinguishes unpredictable shocks, predictable shocks, and sick days.
- Severe shocks are those that cause deterioration in functional health.
- Health shocks may be treated or untreated, and untreated shocks generate worse health transitions.

Interpretation: Health is not modeled as only a medical-cost shock. It is a state variable that changes labor-market opportunities, preferences for work, productivity, and future health risk.

3.3 Insurance and access to treatment

- Employer-sponsored health insurance (ESHI) arrives as part of job offers.

- Medicaid and other means-tested programs provide coverage or transfers for low-income agents.
- Some uninsured agents can access free care, low-cost safety-net care, or can treat and default on bills.
- OOP treatment costs depend on insurance status.
- Lack of ESHI can matter through both higher OOP costs and reduced access to treatment.

Interpretation: The model distinguishes three margins that are often collapsed in simpler models: price of care, access to care, and ability to avoid paying through free care or default.

3.4 Labor-market states and human capital

- Work experience X_t evolves with employment, especially full-time work.
- Wage offers depend on education, experience, health, and lagged employment status.
- Employment offers combine wages, hours, and insurance.
- Lagged employment matters because leaving full-time employment can reduce future full-time offer probabilities.
- Human capital amplifies health shocks because a temporary health shock can lower future work experience.

Interpretation: A health shock is persistent not only because health may deteriorate. It is persistent because the labor market state changes after the shock.

3.5 Key outcome measures

- Present value of lifetime earnings (PVE).
- Present value of lifetime consumption.
- Employment rate and full-time years worked.
- Mean wage offer.
- Fraction in good health and life expectancy.
- Medical treatment, payment sources, and OOP costs.
- Social insurance receipt and government surplus.
- Inequality measures such as the coefficient of variation of PVE.

Interpretation: The paper evaluates health shocks and insurance across both private welfare and public budgets. This is essential because health insurance changes medical spending, work incentives, and transfer reliance simultaneously.

4 Models and theoretical specifications

4.1 State variables and timing

At the beginning of period t , an individual is characterized by assets A_t , experience X_t , functional health H_t , risk factors R_t , marital status M_t , lagged employment/insurance status O_{t-1} , and spouse variables when relevant. The period then proceeds through mortality, job offers, employment choice, health shocks, treatment and payment decisions, consumption/saving, and next-period state transitions.

Interpretation: The ordering matters. Employment and insurance are chosen before health shocks are realized, so insurance is valuable ex ante. Treatment decisions occur after shocks, so the model captures moral hazard and access barriers after illness occurs.

4.2 Employment offers and human capital

A job offer is a tied package:

$$O_t^* = (w_t, h_t, I_t),$$

where w_t is the wage offer, h_t is hours status, and I_t indicates ESHI. Offer probabilities and wage offers depend on lagged employment, education, experience, and health.

Interpretation: This tied-offer structure is important because workers cannot choose insurance independently of employment. Health insurance is part of the job ladder.

4.3 Health production and treatment

Health evolves according to lagged health, health shocks, treatment choices, age, and latent health type. Treatment reduces the probability that a shock causes persistent deterioration:

$$H_{t+1} = g(H_t, d_t, T_t, age_t, type).$$

Interpretation: Treatment is a health investment. The agent weighs current OOP costs and access constraints against better future health, better employment prospects, and higher future earnings.

4.4 Utility, budget, and social insurance

Individuals choose consumption and saving after employment and health decisions. Those unable to finance minimum consumption receive a transfer that proxies for food stamps, disability payments, Medicaid, and other means-tested support.

$$A_{t+1} = R(A_t + y_t + transfers_t - c_t - medical\ payments_t).$$

Interpretation: Means-tested transfers create insurance but can also weaken work incentives for low-skill workers facing high health risk.

4.5 Counterfactual health shock decompositions

The paper studies counterfactuals that shut down channels of health shocks:

1. Labor supply effect: remove direct effects of shocks on work days and labor supply while holding other margins fixed.
2. Human capital effect: allow experience accumulation to change through employment responses.
3. Health productivity effect: allow health status to affect productivity and preferences for work.
4. Behavioral effect: allow agents to re-optimize decision rules when health risk is removed.

Interpretation: The decomposition distinguishes immediate damage from dynamic amplification. The human capital channel is especially important for low-skill workers because employment exits reduce future wage and job opportunities.

4.6 Public insurance counterfactual

The main policy counterfactual gives public insurance to workers who lack ESHI. This changes OOP costs, access to treatment, payment sources, treatment probabilities, health transitions, labor supply, earnings, tax revenues, transfers, and unpaid bills.

Interpretation: The counterfactual is not simply “more medical spending.” It changes the economic environment in which low-skill workers choose whether work is worthwhile.

5 Calibration, validation, and empirical design

5.1 Calibration logic

- Health transitions, spending distributions, and treatment probabilities are disciplined by MEPS.
- Employment, wages, hours, and insurance offer probabilities are disciplined by CPS and related labor-market data.
- Asset accumulation and bequest motives are disciplined by HRS and PSID patterns.
- The model is validated against non-targeted moments, including health gradients and earnings inequality over the life cycle.

Interpretation: The credibility of the paper depends on whether one model can match many observed patterns without mechanically forcing each correlation into the health production function.

5.2 Why the model fit is informative

- Table 1 tests whether the model reproduces empirical correlations among health, education, income, and insurance.
- Figures 2 and 3 test whether the model generates plausible dynamic responses to severe health shocks.
- Tables 3–6 test whether decompositions of earnings, employment, inequality, and transfers line up with the channels in the model.
- Tables 7–9 test whether the policy response to public insurance comes mainly through cost reduction or treatment access.

Interpretation: Because the model is structural, credibility does not come from a clean quasi-experimental policy shock. It comes from matching many moments and then using the model to decompose mechanisms that are hard to isolate in reduced-form data.

5.3 Main limitations

- The model focuses on men. Extending it to women would require fertility, pregnancy costs, and more complex labor-supply dynamics.
- The model abstracts from general equilibrium effects on wages, taxes, and medical prices.
- Health shocks are categorized from survey and diagnostic data, which may underreport true shocks.
- Public insurance counterfactuals depend on structural assumptions about access, moral hazard, free care, and default.

- Racial minority extensions are limited by sample sizes, especially for higher-educated Black and Hispanic men.

Interpretation: These limitations matter, but the paper’s main contribution is the fully specified mechanism linking health, treatment, insurance, human capital, and labor supply.

6 Section-by-section study guide

- **Introduction.** Establishes the key questions: how much health shocks reduce earnings, how much they contribute to inequality, and how insurance modifies those effects.
- **Literature.** Positions the paper as a bridge between health-investment models, medical-expense risk models, and life-cycle human-capital/labor-supply models.
- **Model.** Describes annual timing, state variables, job offers, health shocks, treatment, payment, default, free care, consumption, savings, transfers, and retirement.
- **Data and measurement.** Explains how MEPS/CPS/HRS/PSID moments discipline shocks, health, insurance, wages, employment, assets, and mortality.
- **Calibration.** Shows how the model is parameterized and what moments are targeted.
- **Model fit.** Demonstrates that the model reproduces observed correlations between health, education, income, and insurance.
- **Health shock results.** Quantifies the dynamic impact of health shocks on PVE, full-time employment, wage offers, and inequality.
- **Policy experiments.** Evaluates public insurance for workers without ESHI and decomposes access versus cost effects.
- **Minority results.** Extends the analysis to low-education Black and Hispanic men.
- **Conclusion.** Summarizes the mechanism: health shocks matter most when they push low-skill workers out of work and into means-tested transfers.

7 Tables and figures

7.1 Figure 1: Timeline of Events and Decisions + Table 1: Ordered Logit Regression for Health Outcomes, Ages 25–64, MEPS Data versus Model

Read:

- Figure 1 lays out the timing of one annual period: state variables, death risk, employment offer, offer acceptance, health shock revelation, treatment/payment decision, consumption/saving, and next-period transitions.
- The timeline highlights that insurance status is tied to employment before health shocks are realized.
- Table 1 compares an ordered logit for health outcomes in MEPS data with regressions on simulated data from the model.
- The model reproduces large effects of lagged health, health shocks, ESHI, income quintiles, and education on health.
- When latent health type and true shocks/treatment are controlled for in the simulated data, the education and ESHI coefficients shrink sharply.

Detailed interpretation: The paired visual establishes both the structure of the model and its first validation exercise. Figure 1 shows why the model is dynamic: health shocks occur after job offers, but their consequences feed into future health, experience, employment, and assets. Table 1 then shows that the model can reproduce observed health gradients without putting education directly into the health production function. This matters because the paper wants to explain the education-health correlation as an equilibrium consequence of selection, insurance, income, treatment, and latent health type.

Economic message: The model can match observed health gradients while preserving a structural interpretation of treatment and insurance channels.

Figure 1: Timeline of events and decisions in each period of the model.



Figure 1: Timeline of events and decisions in each period of the model.

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TABLE 1
ORDERED LOGIT REGRESSION FOR HEALTH OUTCOMES, AGES 25–64,
MEPS DATA VERSUS MODEL

	Data	Model 1	Model 2	Model 3	Model 4
Lagged H = poor	−4.483*** (.105)	−4.571*** (.018)	−4.406*** (.019)	−5.374*** (.020)	−5.225*** (.020)
Lagged H = fair	−1.818*** (.041)	−1.709*** (.007)	−1.589*** (.007)	−1.836*** (.007)	−1.708*** (.007)
Some college	.128*** (.049)	.199*** (.008)	.027*** (.008)	.205*** (.008)	.021** (.009)
College	.353*** (.049)	.449*** (.008)	.008 (.009)	.492*** (.009)	.012 (.010)
d^h shock	−.726*** (.057)	−.769*** (.010)	−.802*** (.010)	−.840*** (.010)	−.885*** (.010)
d^w shock	−.675*** (.046)	−.776*** (.008)	−.812*** (.008)	−.923*** (.008)	−.973*** (.008)
ESHI	.446*** (.050)	.588*** (.007)	.643*** (.007)	−.032*** (.008)	.010 (.008)
Including 1st quintile	−.301*** (.064)	−.234*** (.011)	−.037*** (.011)	−.210*** (.011)	.008 (.011)
Including 2nd quintile	−.100* (.061)	−.152*** (.010)	−.029*** (.010)	−.137*** (.010)	−.003 (.010)
Including 4th quintile	.093 (.064)	.099*** (.010)	.039*** (.010)	.073*** (.010)	.005 (.011)
Including 5th quintile	.191*** (.067)	.135*** (.011)	.055*** (.011)	.087*** (.012)	−.004 (.012)
Latent health = bad			−.917*** (.008)		−1.005*** (.008)
Untreated health shock				−1.703*** (.011)	−1.751*** (.011)
Cubic age	Yes	Yes	Yes	Yes	Yes
Shocks correctly measured	No	No	No	Yes	Yes
Pseudo R^2	.271	.263	.278	.305	.322

NOTE.—The table provides information on how well the model fits the health process, as well as the mechanisms that generate observed correlations among health, education, and income in

Table 1: Ordered logit regression for health outcomes, MEPS data versus model.

7.2 Table 2: Effects of Severe Health Shocks on Present Value of Earnings + Figure 2: Effects of a Severe Health Shock at Age 40 on Wages, Employment, and Earnings

Read:

- Table 2 reports the average effect of a severe health shock on remaining lifetime earnings and full-time work.
- At age 40, high-school-or-less men lose about \$55,000, or 11.5% of remaining PVE, more than twice the percentage loss for college men.
- The human capital channel explains a large share of the loss: about 42% for high-school men shocked at age 40 and 34% for college men.
- Figure 2 shows that wage offers recover gradually, but full-time employment recovers much more slowly.
- The gap between the full shock path and the human-capital-fixed counterfactual shows how employment history amplifies the original shock.

Detailed interpretation: The table and figure together show that health shocks are not just short-run medical or sick-day events. The immediate reduction in work reduces experience accumulation and changes future employment offers. The employment panel is especially important: even when wage offers recover, workers who leave full-time employment face persistent reductions in full-time offer probabilities. The human capital share in Table 2 is therefore a measure of dynamic amplification, not just contemporaneous lost earnings.

Economic message: Health shocks are more damaging for low-skill workers because they disrupt future job ladders and human capital accumulation, not merely because they reduce current health.

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Education, Age of Shock	Δ PV Earnings				Δ FT Year Work		
	<i>HC</i> Fixed	Total Effect		% of Total Due to <i>HC</i>	<i>HC</i> Fixed	Total	
	%	%	%				
High school or less:							
30	-20,414	-3.7	-46,520	-8.4	56.1	-.66	-1.97
40	-31,823	-6.7	-55,006	-11.5	42.1	-.91	-1.89
50	-40,812	-12.3	-59,495	-18.0	31.4	-1.05	-1.66
60	-31,485	-30.0	-37,229	-35.4	15.4	-.77	-.91
Some college:							
30	-20,538	-3.1	-50,036	-7.4	59.0	-.55	-1.65
40	-29,055	-5.1	-49,996	-8.7	41.9	-.70	-1.44
50	-33,293	-8.7	-48,837	-12.8	31.8	-.72	-1.18
60	-26,659	-21.8	-32,418	-26.5	17.8	-.55	-.68
College:							
30	-25,641	-2.4	-56,115	-5.2	54.3	-.37	-1.10
40	-35,551	-3.7	-53,859	-5.6	34.0	-.41	-.84
50	-42,045	-6.5	-50,788	-7.9	17.2	-.43	-.63
60	-38,700	-18.1	-46,246	-21.7	16.3	-.45	-.58

NOTE.—The table shows the impact of severe health shocks on lifetime earnings and work years and how this depends on the age when the shock occurs. To do this, we compare the baseline model simulation to counterfactuals where either (i) all men experience a severe d^h shock at the indicated age or (ii) no man experiences such a shock at the indicated age. In the “HC fixed” scenario, we hold human capital fixed at the levels that arise in the baseline

Table 2: Effects of severe health shocks on present value of earnings.

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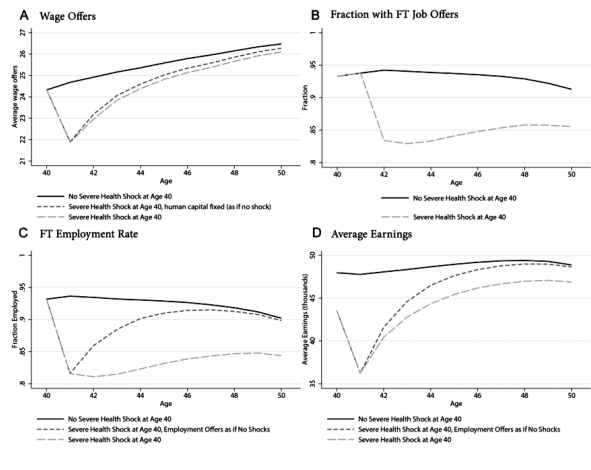


Figure 2: Effects of a severe health shock at age 40 on wages, employment, and earnings.

7.3 Figure 3: Effects of a Severe Health Shock on Full-Time Employment, by Education + Table 3: Effect of Health Shocks on Key Outcomes in Baseline Model

Read:

- Figure 3 decomposes the full-time employment effect by education.
- High-school men suffer the largest and most persistent employment drop; college men have much smaller and shorter-lived effects.
- Table 3 compares the baseline model with counterfactuals that remove shocks or high risk at ages 25–64.
- Eliminating shocks increases survival, good health, employment, years worked, and wage offers.
- The improvements are larger for high-school-or-less men than for all men.

Detailed interpretation: Figure 3 is the cleanest visual showing the education gradient in shock persistence. The low-education group does not simply take a one-year hit; it moves onto a lower employment trajectory. Table 3 generalizes this pattern to several outcomes and shows that eliminating health shocks improves employment and earnings more than reducing predictable high-risk states alone. This reinforces the idea that realized shocks and treatment dynamics, not just baseline risk, drive much of the labor-market damage.

Economic message: Health shocks widen lifetime inequality because low-education workers experience larger and more persistent employment losses.

fixed, as in the baseline (dashed dark-gray line), so that job/wage offers

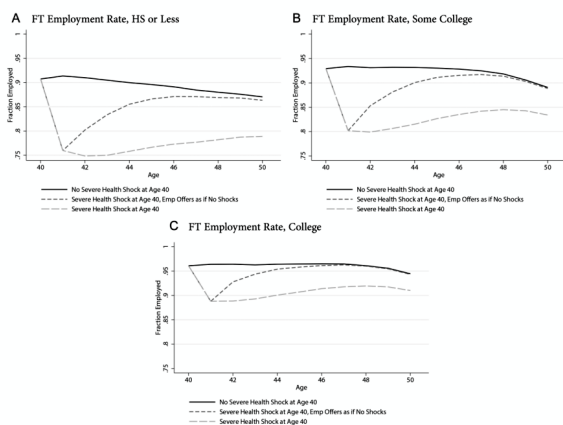


FIG. 3.—Effects of a severe health shock at age 40 on full-time employment, by education. This figure illustrates channels through which severe health shocks affect employment rates. The solid line is a simulation where no one has a d^* shock at age 40. The dashed light-gray line is a simulation where all men have a d^* shock followed by health deterioration from ages 40 to 41. The dashed dark-gray line is the same experiment, except we give agents the same em-

Figure 3: Effects of a severe health shock on full-time employment, by education.

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TABLE 3
EFFECT OF HEALTH SHOCKS ON KEY OUTCOMES IN BASELINE MODEL

	Medical Costs	Sick Days	Survival to 65 (%)	Good H (%)	Emp (%)	Years Worked	SI (%)	Wage Offer
All men (25–64):								
Baseline	4,626	16.42	82.06	59.95	87.74	30.65	6.56	23.60
No s , d^* , d^*	1,127	.00	87.09	74.76	90.99	34.17	2.67	24.22
No high R	4,069	14.35	82.68	61.54	88.41	31.18	5.94	23.68
High school or less:								
Baseline	5,160	22.50	78.04	49.93	84.77	28.52	10.18	17.55
No s , d^* , d^*	1,130	.00	85.22	67.69	89.82	33.56	3.98	18.35
No high R	4,481	19.79	79.07	51.79	85.87	29.27	9.11	17.68

NOTE.—The table summarizes how health shocks affect several key outcomes: medical costs, sick days, etc. The row labeled “baseline” shows summary statistics for data simulated from the baseline model. The rows labeled “no s , d^* , d^* ” and “no high R ” show simulations of counterfactuals where the indicated health shocks (or the possibility of high-risk R) are shut down at ages 25–64 but with decision rules unchanged. Medical costs are equal to $0.6 \cdot \text{Charges}$. Sick days are full-time days per year, assuming a working day of 8 hours.

Table 3: Effect of health shocks on key outcomes in the baseline model.

7.4 Table 4: Effects of Health Shocks on Present Value of Lifetime Earnings + Table 5: Effects of Health Shocks on Inequality in Lifetime Earnings

Read:

- Table 4 decomposes the PVE effect into labor supply, human capital, health productivity, behavioral, and total effects.
- For all men, eliminating health shocks increases PVE by about 10.65%.
- Effects are much larger for low-productivity high-school men: the total PVE gain is 36.61%.
- The human capital and behavioral channels are especially important for low-skill groups.
- Table 5 shows health shocks contribute meaningfully to inequality in PVE: eliminating shocks reduces the coefficient of variation by about 12% for all men.
- The inequality effects are largest among high-school-or-less men.

Detailed interpretation: The decomposition is the paper’s main mechanism table. The labor supply effect is the direct current-work margin. The human capital effect captures lower future experience. The health effect captures the productivity/preference role of health. The behavioral effect captures re-optimization when risk is removed. The behavioral channel is crucial: low-skill workers facing high health risk may rationally reduce work effort and rely more on means-tested transfers. Table 5 then shows why this matters for inequality: health shocks disproportionately harm workers who already have weak attachment and low wages.

Economic message: Health shocks generate inequality through a combination of bad luck and rational behavioral responses to a risky, means-tested environment.

TABLE 4
EFFECTS OF HEALTH SHOCKS ON PRESENT VALUE OF LIFETIME EARNINGS (PVE)

	Baseline PVE	Labor Supply Effect (1)	Human Capital Effect (2)	Health Effect (3)	Behavior Effect (4)	Total Change (5)
All	775,007	3.72	2.75	1.41	.78	10.65
≤High school	562,440	8.75	4.78	1.57	2.80	17.88
Some college	686,287	5.57	3.04	1.02	.22	9.85
College	1,086,707	3.94	1.35	1.48	-.22	6.54
≤High school						
Low productivity	317,861	10.74	14.82	1.26	9.80	36.61
Medium productivity	548,271	9.23	4.26	1.64	2.15	17.28
High productivity	821,246	7.62	1.24	1.65	.52	11.03
Some college						
Low productivity	420,611	6.41	7.15	.90	1.88	16.32
Medium productivity	667,130	5.50	2.66	1.04	-.20	9.00
High productivity	971,121	5.19	1.53	1.06	-.22	7.56
College						
Low productivity	604,205	4.54	3.13	1.47	-.44	8.50
Medium productivity	1,039,861	3.97	.97	1.51	-.17	6.29
High productivity	1,569,256	3.74	.75	1.47	-.15	5.81

NOTE.—The table shows how health shocks affect the present value of lifetime earnings (PVE). The first column presents the mean of the PVE for all men and for different education/skill types, in the baseline model in 2010 dollars. The next five columns show the percentage change of the PVE from the baseline in simulations where various channels through which health shocks affect earnings are shut down: cols. 1–5 present the direct effects of health shocks decomposed into the labor supply effect, the human capital effect, and the health productivity effect. Column 4 presents the behavioral effect (due to decision rules adapting), and col. 5 presents the total effect from a simulation where health risk is eliminated and decision rules adapt to the new environment.

Table 4: Effects of health shocks on present value of lifetime earnings.

TABLE 5
EFFECTS OF HEALTH SHOCKS ON INEQUALITY IN LIFETIME EARNINGS (CV of PVE)

	Baseline CV	Labor Supply Effect (1)	Human Capital Effect (2)	Health Effect (3)	Behavior Effect (4)	Total Change (5)
All	.497	−4.15	−5.23	.01	−2.62	−11.98
≤High school	.454	−4.76	−9.45	.03	−7.32	−21.50
Some college	.395	−3.35	−6.30	−.03	−1.97	−11.65
College	.350	−1.65	−3.13	−.19	.80	−4.17

NOTE.—The table shows how health shocks contribute to lifetime earnings inequality. The first column presents the coefficient of variation (CV) of the present value of lifetime earnings (PVE) for agents in the baseline model. The next five columns show the percentage change from the baseline in simulations where various channels through which health shocks affect earnings are shut down. The experiments in cols. 1–5 are explained in the table 4 note.

The rightmost column of table 4 shows the change in the mean PVE when health shocks are eliminated at working ages.²⁹ Importantly, in

Table 5: Effects of health shocks on inequality in lifetime earnings.

7.5 Table 6: Effects of Health Shocks on Employment and Social Insurance + Table 7: Mandatory Public Health Insurance Impact on Health Spending/Sources of Payment

Read:

- Table 6 shows that eliminating health shocks raises employment and lowers social-insurance receipt.
- For low-skill high-school workers, baseline employment is only 71.5%, but eliminating shocks and allowing full adjustment raises it above 92%.
- Social-insurance receipt falls sharply when health shocks are eliminated, especially for low-skill groups.
- Table 7 shows that public insurance for the uninsured raises treatment spending for the no-ESHI group but sharply reduces untreated shock costs.

- Public insurance shifts payments away from unpaid bills and Medicaid/free care toward the public program.

Detailed interpretation: These tables connect health shocks to public budgets. Table 6 shows that health shocks indirectly raise transfer spending by lowering employment. This is not just a private earnings loss; it is a fiscal channel. Table 7 shows the medical spending side of the policy experiment. Spending rises because more people treat, but the treatment prevents future health deterioration and reduces unpaid bills/free care. Thus, looking only at direct health spending overstates the fiscal cost of insurance.

Economic message: Health insurance has a fiscal trade-off: it raises formal medical spending but reduces untreated-shock costs, unpaid care, and transfer reliance.

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TABLE 6
EFFECTS OF HEALTH SHOCKS ON EMPLOYMENT AND SOCIAL INSURANCE

	Employment (%)				Social Insurance (%)			
	Baseline	No Health Shocks			Baseline	No Health Shocks		
		LS Effect	LS + HC Effect	Total Effect		LS Effect	LS + HC Effect	Total Effect
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
All	87.7	88.6	91.0	92.4	6.6	4.6	2.7	1.5
≤High school	84.8	86.2	89.8	93.0	10.2	7.0	4.0	1.4
Some college	86.8	87.7	90.1	90.7	6.6	4.8	2.9	2.1
College	91.7	92.0	92.9	92.7	2.4	1.7	1.0	1.2
≤High school:								
Low skill	71.5	74.7	84.8	92.3	25.5	19.1	10.3	2.7
Medium skill	89.5	91.1	91.9	93.4	4.0	1.3	1.0	.6
High skill	93.2	92.7	92.8	93.5	1.1	.7	.7	.7
Some college:								
Low skill	79.3	80.8	86.3	88.6	15.7	12.5	7.3	4.6
Medium skill	89.8	90.4	91.6	91.3	2.8	1.4	.8	.9
High skill	91.3	91.7	92.2	92.0	1.3	.6	.4	.7
College:								
Low skill	88.5	89.2	91.5	91.1	6.0	4.3	2.5	2.7
Medium skill	92.8	92.9	93.3	93.0	.8	.5	.4	.5
High skill	93.9	93.9	94.1	93.9	.3	.2	.1	.3

NOTE.—The table shows how health shocks affect full-time employment and government transfer receipt of working-age men. Column 1 shows rates of employment or transfer re-

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TABLE 7
MANDATORY PUBLIC HEALTH INSURANCE IMPACT ON HEALTH SPENDING/SOURCES OF PAYMENT

	Total Cost of Treatment	Sources of Payment				Untreated Shocks
		OOP (Self)	ESHI	Medicaid	Unpaid Public	
Baseline:						
No ESHI	2,556	506		937	1,113	1,624
ESHI	2,606	522	2,043	1	39	32
All	2,586	515	1,234	372	465	662
Public insurance:						
No ESHI	3,966	517		105	61	3,282
ESHI	2,550	517	1,994	1	39	31
All	3,114	517	1,200	42	48	1,307

NOTE.—The top half shows the average (or per capita) annual cost of treatment for health shocks, and the sources of payments for those costs, in the baseline model. We report this for all men, as well as for those with and without ESHI. The sources of payment are OOP (self), ESHI, and Medicaid, with the remainder being unpaid bills. Finally, the last column shows the latent cost of untreated health shocks. The bottom half shows how costs and sources of payment change when public insurance is provided to men who lack ESHI in the baseline. We report this for all men and for those with and without ESHI in the baseline.

Table 6: Effects of health shocks on employment and social insurance.

Table 7: Mandatory public health insurance impact on health spending/sources of payment.

7.6 Table 8: Effects of Providing Mandatory Public Health Insurance + Table 9: How Does Public Insurance Affect Health and Earnings? Mechanisms

Read:

- Table 8 shows that public insurance raises treatment conditional on shock from 80.1% to 99.2%.
- The fraction in good health rises from 60.0% to 65.9%, employment rises from 87.7% to 88.7%, and PVE rises from 774 to 789 thousand dollars.
- Government surplus for working-age male-headed households changes little because higher program costs are partly offset by higher taxes, lower social insurance, and lower unpaid bills.
- Table 9 compares public insurance to two mechanisms: zero OOP cost and full access to treatment.
- Zero OOP cost alone has small effects, while making treatment accessible produces effects close to public insurance.

Detailed interpretation: Table 8 is the main policy table. It shows the program is costly in medical spending terms but much less costly once indirect effects are counted. Table 9 is the key mechanism table for the policy. The fact that “everyone can treat” gets close to the public-insurance outcome implies that

access, not just price, is the binding margin for uninsured workers. Zero OOP costs without access do little because many uninsured workers cannot obtain care even when willingness to pay is high.

Economic message: Public insurance matters primarily because it grants access to treatment, not simply because it reduces OOP prices.

TABLE 8
EFFECTS OF PROVIDING MANDATORY PUBLIC HEALTH INSURANCE

	Baseline	Public HI	Change
Average annual medical expenses (per man), ages 25–64:			
Costs covered by new public insurance program	0	1,307	
OOP costs, paid by individuals	515	517	
Costs covered by ESHI	1,234	1,200	
Costs covered by Medicaid/DI	372	42	
Unpaid bills	465	48	
Total (sum of all above expenditures)	2,586	3,114	528
Average costs of untreated health shocks	662	42	
Health outcomes:			
Fraction who treat conditional on shock, ages 25–64	80.1	99.2	
Fraction experiencing a health shock, ages 25–64	59.9	58.4	
Fraction in good health, ages 25–64	60.0	65.9	
Life expectancy	77.5	78.2	
Labor market outcomes:			
Mean wage offer	23.60	23.82	
Employment rate	87.7	88.7	
FT years of work	30.64	31.39	
PV of lifetime earnings (thousands)	774	789	
Coefficient of variation PV earnings	.497	.481	
PV of lifetime consumption (thousands)	800	810	
Coefficient of variation PV consumption	.425	.415	
Receive social insurance benefits, working ages (%)	6.6	5.4	
Government revenues and expenditures (per household):			
Households with working-age male heads only:			
Tax revenues	15,403	15,530	
Public health insurance premiums	0	323	
Public health insurance payments	0	–1307	
Social insurance payments	–953	–541	
Unpaid medical bills	–465	–48	
Government surplus (per household), head age 25–64	13,985	13,957	–28
Households with heads of all ages (all model households):			
Tax revenues	12,901	12,962	
Social security and Medicare (net of Medicare premiums)	–10,630	–10,834	
Social insurance payments (other)	–790	–477	
Public health insurance (payments minus premiums)	0	–708	
Unpaid bills	–336	–34	
Government surplus (per household), all ages	1,145	908	–237

NOTE.—This table shows how several important outcomes change between the baseline and a counterfactual where we provide public insurance to men who lack ESHI in the baseline. The first column shows baseline outcomes, the second column shows counterfactual outcomes, and the third column shows changes in key expenditure variables.

Table 8: Effects of providing mandatory public health insurance.

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TABLE 9
HOW DOES PUBLIC INSURANCE AFFECT HEALTH AND EARNINGS? MECHANISMS

HOW DOES PUBLIC INSURANCE AFFECT HEALTH AND EARNINGS: MECHANISMS							
	Treat	Pay	Good H	Employed	SI	PVE	CV
All men (25–64):							
Baseline	80.1	86.2	60.0	87.7	6.6	774	.497
Public insurance	99.2	98.3	65.9	88.7	5.4	789	.481
Zero OOP cost	81.7	60.4	88.4	88.4	5.3	777	.489
Everyone can treat	96.6	89.0	65.2	88.2	6.1	785	.490
High school or less (25–64):							
Baseline	77.4	84.0	49.9	84.8	10.2	562	.454
Public insurance	98.6	97.1	56.9	86.8	8.0	584	.424
Zero OOP cost	79.9	50.6	85.9	85.9	8.2	569	.437
Everyone can treat	94.7	87.2	55.7	85.9	9.1	577	.441

NOTE.—The table shows impact of three experiments on key outcomes. The experiments are (i) providing public insurance to the uninsured, (ii) eliminating the OOP cost of health shocks, and (iii) giving 100% access to care to the uninsured, provided they are willing to pay. The percentage who treat is conditional on having a health shock. The percentage who pay is conditional on having a shock and treating. PV earnings are in thousands of dollars. The last column presents the coefficient of variation of PV earnings. All statistics are calculated in the regression of individual head 25–64.

Table 9: How does public insurance affect health and earnings? Mechanisms.

7.7 Table 10: Effects of Health Shocks on Present Value of Lifetime Earnings, Racial Minorities

Read:

- Table 10 applies the decomposition to Black and Hispanic men with high school or less education.
- Health shocks reduce PVE for both groups through labor supply, human capital, health productivity, and behavior.
- Effects are especially large for low-productivity Black men: eliminating health shocks raises PVE by 77.29%.
- The human capital effect for low-productivity Black men is 36.71%, and the behavioral effect is 24.35%.
- Hispanic men show qualitatively similar but smaller effects than Black men.

Detailed interpretation: This table shows that the core mechanism generalizes beyond the White male calibration. The strongest effects occur for workers with the weakest baseline labor-market position. Health shocks do not only reduce productivity; they push low-wage workers out of the labor market, making it harder to rebuild experience and increasing reliance on transfers. The minority results also emphasize that policy conclusions depend on access to care and group-specific baseline health and labor-market conditions.

Economic message: The burden of health risk is largest where low wages, weak employment attachment, and poor access to care interact.

TABLE 10
EFFECTS OF HEALTH SHOCKS ON PRESENT VALUE OF LIFETIME EARNINGS
(PVE), RACIAL MINORITIES

	Baseline PVE	Labor Supply Effect (1)	Human Capital Effect (2)	Health Effect (3)	Behavior Effect (4)	Total Change (5)
Black men $\leq$ high school	421,798	9.24	8.10	1.57	4.80	23.71
Low productivity	167,139	14.73	36.71	1.50	24.35	77.29
Medium productivity	412,442	9.80	7.16	1.49	3.92	22.37
High productivity	685,875	7.57	1.70	1.63	.57	11.46
Hispanic men $\leq$ high school	482,608	8.70	4.98	1.77	2.28	17.73
Low productivity	244,637	10.58	16.44	1.70	7.11	35.83
Medium productivity	496,821	8.42	3.88	1.72	1.76	15.77
High productivity	706,425	8.23	1.79	1.84	.97	12.83

NOTE.—The table shows how health shocks affect the present value of lifetime earnings (PVE) for Black and Hispanic men with high school or less education. The first column presents the mean of the PVE for all Black or Hispanic men with $\leq$ high school education and for different skill types in the baseline model in 2010 dollars. The next five columns show the percentage change of the PVE from the baseline in simulations where various

Table 10: Effects of health shocks on present value of lifetime earnings, racial minorities.

8 Result map

- **Model fit result.** The model reproduces observed education-income-insurance-health correlations without needing direct education effects in health production.
- **Shock effect result.** Severe shocks reduce lifetime earnings substantially, especially for low-education workers.
- **Human capital amplification result.** A large share of the earnings loss comes from reduced future work experience and employment offers.
- **Inequality result.** Health shocks account for a nontrivial share of lifetime earnings inequality, especially within low-education groups.
- **Behavioral result.** Health risk changes decision rules; low-skill workers may rationally rely more on transfers because shocks can push them to the consumption floor.
- **Insurance access result.** Public insurance is valuable largely because it increases access to treatment, not only because it lowers treatment prices.
- **Fiscal result.** Public insurance raises direct spending but reduces transfer reliance and unpaid care, so the net program cost is modest in the model.
- **Minority result.** Black and Hispanic men display qualitatively similar channels, with especially large effects for low-productivity Black men.

9 Short summary

- The paper builds a dynamic life-cycle model linking health shocks, health insurance, treatment, human capital, labor supply, earnings, and transfers.
- Health shocks have persistent effects because they reduce employment, slow human capital accumulation, lower future job offers, and worsen health.
- The model matches many observed correlations in the data, including health gradients by education, income, and insurance.
- Severe health shocks cause large losses in present value of earnings, especially for less-educated workers.
- Human capital is a major amplification mechanism: health shocks lead to employment exits, and employment exits reduce future wage offers.
- Health shocks contribute materially to lifetime earnings inequality, particularly within low-skill groups.
- Insurance matters through access to treatment, not just through lower OOP costs.
- Providing public insurance to uninsured men increases treatment and medical spending but improves health, employment, earnings, and reduces social-insurance reliance.
- The policy has a modest net budget cost in the model because higher direct public insurance spending is partly offset by lower unpaid bills, lower Medicaid/free care, higher taxes, and lower transfer payments.
- The mechanisms are similar for White, Black, and Hispanic men, but the magnitude is larger for low-skill Black men because baseline employment and earnings are lower.

10 Conclusion

10.1 Core conclusion

The paper’s central conclusion is that health shocks are a major dynamic force behind earnings losses and earnings inequality. Their effects operate not only through current medical costs or sick days but through

labor-market detachment, human capital loss, lower future job offers, lower future wages, and behavioral responses to means-tested social insurance.

10.2 Economic intuition

A worker who experiences a health shock may leave full-time work, lose experience, face lower future offer probabilities, and become more likely to rely on transfers. This effect is much more severe for low-skill workers because their wage offers are lower and because means-tested programs provide an outside option that can make labor-market re-entry less attractive. Health shocks therefore create inequality through both realized bad luck and anticipatory behavioral responses to risk.

10.3 Why insurance has multiple roles

The paper shows that insurance is not merely a consumption-smoothing tool. It changes whether treatment is available, whether shocks become severe, whether future health deteriorates, whether workers remain attached to the labor force, and whether low-skill workers rely on social insurance. The main policy counterfactual demonstrates that access to treatment is at least as important as OOP cost reductions.

10.4 Policy implications

The public-insurance counterfactual implies that non-means-tested coverage for the uninsured can improve health and labor-market outcomes while reducing reliance on means-tested transfers. This is very different from Medicaid-style insurance, which can provide health protection but may also discourage labor supply when eligibility depends on low income. The paper’s policy message is that the design of insurance matters as much as the generosity of insurance.

10.5 Methodological contribution

The paper demonstrates the value of structural life-cycle modeling for health and earnings questions. A reduced-form estimate of a health shock’s effect may miss selection into shocks and treatment, and it may not decompose labor supply, human capital, health productivity, and behavioral mechanisms. A structural model can compare counterfactual environments that are not directly observed, such as public insurance for all workers without ESHI.

10.6 Limitations and interpretation

The results depend on the model’s treatment of health shocks, access to care, free care, default, transfers, offer probabilities, and health transitions. The analysis focuses on men, and extending the framework to women would require modeling fertility, pregnancy, and richer family labor supply. The paper also abstracts from general equilibrium effects on taxes, wages, medical prices, and employer insurance offers. These limitations are important, but they do not change the main insight that health and labor-market dynamics are tightly linked.

10.7 Takeaway

Health shocks are labor-market shocks. They damage lifetime earnings by reducing work, weakening human capital accumulation, lowering future job offers, and changing behavior around public transfers. Insurance that improves treatment access can therefore be an earnings policy, an inequality policy, and a public-budget policy, not merely a medical-spending policy.